[Paper 6 continued from p. 28.]

The following expressions are obtained:

$$\begin{aligned}
\alpha &= \cos^2 f + \sin^2 f \cos\theta,\\
\alpha' &= \cos f \cos g + \sin f \sin g \cos(z-\theta),\\
\alpha'' &= \cos f \cos h + \sin f \sin h \cos(y+\theta),\\
\beta &= \cos g \cos f + \sin g \sin f \cos(z+\theta),\\
\beta' &= \cos^2 g + \sin^2 g \cos\theta,\\
\beta'' &= \cos g \cos h + \sin g \sin h \cos(x-\theta),\\
\gamma &= \cos h \cos f + \sin h \sin f \cos(y-\theta),\\
\gamma' &= \cos h \cos g + \sin h \sin g \cos(x+\theta),\\
\gamma'' &= \cos^2 h + \sin^2 h \cos\theta.
\end{aligned}$$

Also

$$\begin{aligned}
\sin g \sin h \cos x &= -\cos g \cos h,\\
\sin h \sin f \cos y &= -\cos h \cos f,\\
\sin f \sin g \cos z &= -\cos f \cos g,
\end{aligned}$$

and

$$\begin{aligned}
\sin g \sin h \sin x &= \cos f,\\
\sin h \sin f \sin y &= \cos g,\\
\sin f \sin g \sin z &= \cos h.
\end{aligned}$$

Substituting,

$$\begin{aligned}
\alpha &= \cos^2 f + \sin^2 f \cos\theta,\\
\alpha' &= \cos f \cos g\,(1-\cos\theta) + \cos h \sin\theta,\\
\alpha'' &= \cos f \cos h\,(1-\cos\theta) - \cos g \sin\theta,\\
\beta &= \cos g \cos f\,(1-\cos\theta) - \cos h \sin\theta,\\
\beta' &= \cos^2 g + \sin^2 g \cos\theta,\\
\beta'' &= \cos g \cos h\,(1-\cos\theta) + \cos f \sin\theta,\\
\gamma &= \cos h \cos f\,(1-\cos\theta) + \cos g \sin\theta,\\
\gamma' &= \cos h \cos g\,(1-\cos\theta) - \cos f \sin\theta,\\
\gamma'' &= \cos^2 h + \sin^2 h \cos\theta.
\end{aligned}$$

Assume $\lambda = \tan\frac{1}{2}\theta\cos f$, $\mu = \tan\frac{1}{2}\theta\cos g$, $\nu = \tan\frac{1}{2}\theta\cos h$, and $\sec^2\frac{1}{2}\theta = 1+\lambda^2+\mu^2+\nu^2 = \kappa$; then

$$\begin{array}{lll}
\kappa\alpha = 1+\lambda^2-\mu^2-\nu^2, & \kappa\alpha' = 2(\lambda\mu+\nu), & \kappa\alpha'' = 2(\nu\lambda-\mu),\\
\kappa\beta = 2(\lambda\mu-\nu), & \kappa\beta' = 1+\mu^2-\nu^2-\lambda^2, & \kappa\beta'' = 2(\mu\nu+\lambda),\\
\kappa\gamma = 2(\nu\lambda+\mu), & \kappa\gamma' = 2(\mu\nu-\lambda), & \kappa\gamma'' = 1+\nu^2-\lambda^2-\mu^2;
\end{array}$$

which are the formulæ required, differing only from those in Liouville, by having λ, μ, ν, instead of $\frac{1}{2}m$, $\frac{1}{2}n$, $\frac{1}{2}p$; and α, α', α''; β, β', β''; γ, γ', γ'', instead of a, b, c; a', b', c'; a'', b'', c''. It is to be remarked, that β', β'', β; γ'', γ, γ', are deduced from α, α', α'', by writing μ, ν, λ; ν, λ, μ, for λ, μ, ν.

Let $1+\alpha+\beta'+\gamma'' = v$; then $\kappa v = 4$, and we have

$$\lambda v = \beta''-\gamma', \qquad \mu v = \gamma-\alpha'', \qquad \nu v = \alpha'-\beta,$$

$$\lambda^2 v = 1+\alpha-\beta'-\gamma'', \qquad v-v = 1-\alpha-\beta'-\gamma''.$$

Suppose that Ax, Ay, Az, are referred to axes $A\text{x}$, $A\text{y}$, $A\text{z}$, by the quantities l, m, n, k, analogous to λ, μ, ν, κ, these latter axes being referred to Ax_1, Ay_1, Az_1, by the quantities l_1, m_1, n_1, k_1.

Let a, b, c; a', b', c'; a'', b'', c''; a_1, b_1, c_1; a_1', b_1', c_1'; a_1'', b_1'', c_1'' denote the quantities analogous to α, β, γ; α', β', γ'; α'', β'', γ''. Then we have, by spherical trigonometry, the formulæ

$$\begin{aligned}
\alpha &= a\,a_1 + b\,a_1' + c\,a_1'', & \beta &= a\,b_1 + b\,b_1' + c\,b_1'', & \gamma &= a\,c_1 + b\,c_1' + c\,c_1'';\\
\alpha' &= a'\,a_1 + b'\,a_1' + c'\,a_1'', & \beta' &= a'\,b_1 + b'\,b_1' + c'\,b_1'', & \gamma' &= a'\,c_1 + b'\,c_1' + c'\,c_1'';\\
\alpha'' &= a''\,a_1 + b''\,a_1' + c''\,a_1'', & \beta'' &= a''\,b_1 + b''\,b_1' + c''\,b_1'', & \gamma'' &= a''\,c_1 + b''\,c_1' + c''\,c_1''.
\end{aligned}$$

Then expressing a, b, c; a', b', c'; a'', b'', c''; a_1, b_1, c_1; a_1', b_1', c_1'; a_1'', b_1'', c_1'', in terms of l, m, n; l_1, m_1, n_1, after some reductions we arrive at

$$\begin{aligned}
kk_1\upsilon &= 4(1 - ll_1 - mm_1 - nn_1) = 4\,\Pi \text{ suppose},\\
kk_1(\beta'' - \gamma') &= 4(l + l_1 + n_1 m - nm_1)\,\Pi,\\
kk_1(\gamma - \alpha'') &= 4(m + m_1 + l_1 n - ln_1)\,\Pi,\\
kk_1(\alpha' - \beta) &= 4(n + n_1 + m_1 n - mn_1)\,\Pi;
\end{aligned}$$

and hence

$$\begin{aligned}
\Pi &= 1 - ll_1 - mm_1 - nn_1, & \Pi\lambda &= l + l_1 + n_1 m - nm_1,\\
\Pi\mu &= m + m_1 + l_1 n - ln_1, & \Pi\nu &= n + n_1 + m_1 n - mn_1,
\end{aligned}$$

which are formulæ of considerable elegance for exhibiting the combined effect of successive displacements of the axes. The following analogous ones are readily obtained:

$$\begin{aligned}
P &= 1 + \lambda l + \mu m + \nu n, & Pl_1 &= \lambda - l - \nu m + \mu n,\\
Pm_1 &= \mu - m - \lambda n + \nu l, & Pn_1 &= \nu - n - \mu l + \lambda m;
\end{aligned}$$

and again,

$$\begin{aligned}
P_1 &= 1 + \lambda l_1 + \mu m_1 + \nu n_1, & P_1 l &= \lambda - l_1 + \nu m_1 - \mu n_1,\\
P_1 m &= \mu - m_1 + \lambda n_1 - \nu l_1, & P_1 n &= \nu - n_1 + \mu l_1 - \lambda m_1.
\end{aligned}$$

These formulæ will be found useful in the integration of the equations of rotation of a solid body.

Next it is required to express the quantities p, q, r, in terms of λ, μ, ν, where as usual

$$\begin{aligned}
p &= \alpha\frac{d\beta}{dt} + \alpha'\frac{d\beta'}{dt} + \alpha''\frac{d\beta''}{dt},\\
q &= \alpha\frac{d\gamma}{dt} + \alpha'\frac{d\gamma'}{dt} + \alpha''\frac{d\gamma''}{dt},\\
r &= \beta\frac{d\alpha}{dt} + \beta'\frac{d\alpha'}{dt} + \beta''\frac{d\alpha''}{dt}.
\end{aligned}$$

Differentiating the values of $\beta\kappa$, $\beta'\kappa$, $\beta''\kappa$, multiplying by γ, γ', γ'', and adding,

$$\kappa p = 2\lambda'(\gamma\mu - \gamma'\lambda + \gamma'') + 2\mu'(\gamma\lambda - \gamma'\mu + \gamma''\nu) + 2\nu'(-\gamma - \gamma'\nu + \gamma''\mu).$$

where λ', μ', ν', denote $\dfrac{d\lambda}{dt}$, $\dfrac{d\mu}{dt}$, $\dfrac{d\nu}{dt}$. Reducing, we have

$$\kappa p = 2(\lambda' + \nu\mu' - \mu'\nu),$$

from which it is easy to derive the system

$$\begin{aligned}
\kappa p &= 2(\lambda' + \nu\mu' - \nu'\mu),\\
\kappa q &= 2(-\nu\lambda' + \mu' + \nu'\lambda),\\
\kappa r &= 2(\mu\lambda' - \lambda\mu' + \nu');
\end{aligned}$$

or, determining λ', μ', ν', from these equations, the equivalent system

$$\begin{aligned}
2\lambda' &= (1 + \lambda^2)p + (\lambda\mu - \nu)q + (\nu\lambda + \mu)r,\\
2\mu' &= (\lambda\mu + \nu)p + (1 + \mu^2)q + (\mu\nu - \lambda)r,\\
2\nu' &= (\nu\lambda - \mu)p + (\mu\nu + \lambda)q + (1 + \nu^2)r.
\end{aligned}$$

The following equation also is immediately obtained,

$$\kappa' = \kappa(\lambda p + \mu q + \nu r).$$

The subsequent part of the problem requires the knowledge of the differential coefficients of p, q, r, with respect to λ, μ, ν; λ', μ', ν'. It will be sufficient to write down the six

$$\kappa\frac{dp}{d\lambda} = 2, \qquad \kappa\frac{dp}{d\lambda'} = 2p\lambda = 0,$$

$$\kappa\frac{dq}{d\lambda} = -2\nu, \qquad \kappa\frac{dq}{d\lambda'} = 2\nu',$$

$$\kappa\frac{dr}{d\lambda} = 2\mu, \qquad \kappa\frac{dr}{d\lambda'} = 2\nu\lambda - 2\mu',$$

from which the others are immediately obtained.

Suppose now a solid body acted on by any forces, and revolving round a fixed point. The equations of motion are

$$\frac{d}{dt}\frac{dT}{d\lambda'} - \frac{dT}{d\lambda} = \frac{dV}{d\lambda},$$
$$\frac{d}{dt}\frac{dT}{d\mu'} - \frac{dT}{d\mu} = \frac{dV}{d\mu},$$
$$\frac{d}{dt}\frac{dT}{d\nu'} - \frac{dT}{d\nu} = \frac{dV}{d\nu},$$

where

$$T = \tfrac{1}{2}(Ap^2 + Bq^2 + Cr^2); \qquad V = \Sigma\left[\int (X dx + Y dy + Z dz)\right] dm;$$

or if $X dx + Y dy + Z dz$ is not an exact differential, $\frac{dV}{d\lambda}$, $\frac{dV}{d\mu}$, $\frac{dV}{d\nu}$, are independent symbols standing for

$$\Sigma\left(X\frac{dx}{d\lambda} + Y\frac{dy}{d\lambda} + Z\frac{dz}{d\lambda}\right) dm, \ldots$$

see Mécanique Analytique, Avertissement, t. I. p. v. [Ed. 3, p. vii.]: only in this latter case V stands for the disturbing function, the principal forces vanishing.

Now, considering the first of the above equations

$$\frac{dT}{d\lambda'} = Ap\frac{dp}{d\lambda'} + Bq\frac{dq}{d\lambda'} + Cr\frac{dr}{d\lambda'} = \frac{2}{\kappa}(Ap - \nu Bq + \mu Cr);$$

whence, writing p', q', r', κ', for $\frac{dp}{dt}$, $\frac{dq}{dt}$, $\frac{dr}{dt}$, $\frac{d\kappa}{dt}$,

$$\frac{d}{dt}\frac{dT}{d\lambda'} = \frac{2}{\kappa}(Ap' - \nu Bq' + \mu Cr') - \frac{2}{\kappa}Bq\nu' + \frac{2}{\kappa}Cr\mu' - \frac{2\kappa'}{\kappa^2}(Ap - \nu Bq + \mu Cr).$$

Also

$$\frac{dT}{d\lambda} = Ap\frac{dp}{d\lambda} + Bq\frac{dq}{d\lambda} + Cr\frac{dr}{d\lambda} = -\frac{2\lambda}{\kappa}(Ap^2 + Bq^2 + Cr^2) + \frac{2}{\kappa}Bq\nu - \frac{2}{\kappa}Cr\mu;$$

and hence $\frac{d}{dt}\frac{dT}{d\lambda'} - \frac{dT}{d\lambda}$

$$= \frac{2}{\kappa}(Ap' - \nu Bq' + \mu Cr') - \frac{2}{\kappa}Bq\nu' + \frac{2}{\kappa}Cr\mu' + \frac{2\lambda}{\kappa}(Ap^2 + Bq^2 + Cr^2) - \frac{2\kappa'}{\kappa^2}(Ap - \nu Bq + \mu Cr).$$

Substituting for λ', μ', ν', κ', after all reductions,

$$\tfrac{1}{2}\left(\frac{d}{dt}\frac{dT}{d\lambda'} - \frac{dT}{d\lambda}\right) = \frac{2}{\kappa}\left[\,\{Ap' + (C-B)qr\} - \nu\{Bq' + (A-C)rp\} + \mu\{Cr' + (B-A)pq\}\,\right];$$

and, forming the analogous quantities in μ, ν, and substituting in the equations of motion, these become

$$\{Ap' + (C-B)qr\} - \nu\{Bq' + (A-C)rp\} + \mu\{Cr' + (B-A)pq\} = \tfrac{1}{2}\kappa\frac{dV}{d\lambda},$$

$$\nu\{Ap' + (C-B)qr\} + \{Bq' + (A-C)rp\} - \lambda\{Cr' + (B-A)pq\} = \tfrac{1}{2}\kappa\frac{dV}{d\mu},$$

$$-\mu\{Ap' + (C-B)qr\} + \lambda\{Bq' + (A-C)rp\} + \{Cr' + (B-A)pq\} = \tfrac{1}{2}\kappa\frac{dV}{d\nu};$$

or eliminating, and replacing p', q', r', by $\dfrac{dp}{dt}$, $\dfrac{dq}{dt}$, $\dfrac{dr}{dt}$, we obtain

$$A\frac{dp}{dt} + (C-B)qr = \tfrac{1}{2}\left\{(1+\lambda^2)\frac{dV}{d\lambda} + (\lambda\mu - \nu)\frac{dV}{d\mu} + (\nu\lambda + \mu)\frac{dV}{d\nu}\right\},$$

$$B\frac{dq}{dt} + (A-C)rp = \tfrac{1}{2}\left\{(\lambda\mu + \nu)\frac{dV}{d\lambda} + (1+\mu^2)\frac{dV}{d\mu} + (\mu\nu - \lambda)\frac{dV}{d\nu}\right\},$$

$$C\frac{dr}{dt} + (B-A)pq = \tfrac{1}{2}\left\{(\nu\lambda - \mu)\frac{dV}{d\lambda} + (\mu\nu + \lambda)\frac{dV}{d\mu} + (1+\nu^2)\frac{dV}{d\nu}\right\};$$

to which are to be joined

$$\kappa p = 2\left(\frac{d\lambda}{dt} + \nu\frac{d\mu}{dt} - \mu\frac{d\nu}{dt}\right),$$

$$\kappa q = 2\left(-\nu\frac{d\lambda}{dt} + \frac{d\mu}{dt} + \lambda\frac{d\nu}{dt}\right),$$

$$\kappa r = 2\left(\mu\frac{d\lambda}{dt} - \lambda\frac{d\mu}{dt} + \frac{d\nu}{dt}\right),$$

where it will be recollected

$$\kappa = 1 + \lambda^2 + \mu^2 + \nu^2;$$

and on the integration of these six equations depends the complete determination of the motion.

If we neglect the terms depending on V, the first three equations may be integrated in the form

$$p^2 = p_1^2 - \frac{C-B}{A}\phi, \qquad q^2 = q_1^2 - \frac{A-C}{B}\phi, \qquad r^2 = r_1^2 - \frac{B-A}{C}\phi,$$

$$2t = \int \frac{d\phi}{\sqrt{\left\{p_1^2 - \dfrac{C-B}{A}\phi\right\}\left\{q_1^2 - \dfrac{A-C}{B}\phi\right\}\left\{r_1^2 - \dfrac{B-A}{C}\phi\right\}}},$$

and considering p, q, r as functions of ϕ, given by these equations, the three latter ones take the form

$$\frac{\kappa}{4qr} = \frac{d\lambda}{d\phi} + \nu\frac{d\mu}{d\phi} - \mu\frac{d\nu}{d\phi},$$

$$\frac{\kappa}{4rp} = -\nu\frac{d\lambda}{d\phi} + \frac{d\mu}{d\phi} + \lambda\frac{d\nu}{d\phi},$$

$$\frac{\kappa}{4pq} = \mu\frac{d\lambda}{d\phi} - \lambda\frac{d\mu}{d\phi} + \frac{d\nu}{d\phi},$$

of which, as is well known, the equations following, equivalent to two independent equations, are integrals,

$$\kappa g = Ap(1 + \lambda^2 - \mu^2 - \nu^2) + 2Bq(\lambda\mu - \nu) + 2Cr(\nu\lambda + \mu),$$

$$\kappa g' = 2Ap(\lambda\mu + \nu) + Bq(1 + \mu^2 - \lambda^2 - \nu^2) + 2Cr(\mu\nu - \lambda),$$

$$\kappa g'' = 2Ap(\nu\lambda - \mu) + 2Bq(\mu\nu + \lambda) + Cr(1 + \nu^2 - \lambda^2 - \mu^2);$$

where g, g', g'', are arbitrary constants satisfying

$$g^2 + g'^2 + g''^2 = A^2p_1^2 + B^2q_1^2 + C^2r_1^2.$$

To obtain another integral, it is apparently necessary, as in the ordinary theory, to revert to the consideration of the invariable plane. Suppose $g' = 0$, $g'' = 0$, then $g = \sqrt{(A^2p_1^2 + B^2q_1^2 + C^2r_1^2)}$, $= k$ suppose.

We easily obtain, where λ_0, μ_0, ν_0, κ_0 are written for λ, μ, ν, κ, to denote this particular supposition,

$$\kappa_0 Ap = 2(\nu_0\lambda_0 - \mu_0)\,k,$$

$$\kappa_0 Bq = 2(\mu_0\nu_0 + \lambda_0)\,k,$$

$$\kappa_0 Cr = (1 + \nu_0^2 - \lambda_0^2 - \mu_0^2)\,k;$$

whence, and from $\kappa_0 = 1 + \lambda_0^2 + \mu_0^2 + \nu_0^2$, $\kappa_0 Cr = (2 + 2\nu_0^2 - \kappa_0)\,k$, we obtain

$$\lambda_0 = \frac{(\nu_0\lambda_0 - \mu_0)\,k}{k + Cr}, \qquad \mu_0 = \frac{(\mu_0\nu_0 + \lambda_0)\,k}{k + Cr}, \qquad \kappa_0 = \frac{(1 + \nu_0^2)\,2k}{k + Cr}.$$

Hence, writing $h = Ap_1^2 + Bq_1^2 + Cr_1^2$, the equation

$$\frac{\kappa_0}{4pq} = \frac{4}{k + Cr}\{(\nu_0\lambda_0 - \mu_0)p + (\mu_0\nu_0 + \lambda_0)q + (1 + \nu_0^2)r\}$$

reduces itself to

$$\frac{4}{k + Cr} = \frac{4\,d\nu_0}{A + Br} \quad \text{i.e.} \quad \frac{4\,d\nu_0}{(k + Cr)\,pq}.$$

The integral takes rather a simpler form if p, q, ϕ be considered functions of r, and becomes

$$2\tan^{-1}\nu_0 = \int \frac{(k + Cr)\,dr}{\sqrt{[k^2 - Bh + (B - C)Cr^2]\,[Ah - k^2 + (C - A)Cr^2]}},$$

and then, ν_0 being determined, λ_0, μ_0, are given by the equations

$$\lambda_0 = \frac{\nu_0 Ap + Bq}{k + Cr}, \qquad \mu_0 = \frac{\nu_0 Bq - Ap}{k + Cr}.$$

Hence l, m, n, denoting arbitrary constants, the general values of λ, μ, ν, are given by the equations

$$P_0\lambda = l + \lambda_0 + m\nu_0 - n\mu_0,$$

$$P_0\mu = m + \mu_0 + n\lambda_0 - l\nu_0,$$

$$P_0\nu = n + \nu_0 + l\mu_0 - m\lambda_0.$$

In a following paper I propose to develope the formulæ for the variations of the arbitrary constants p_1, q_1, r_1, l, m, n, when the terms involving V are taken into account.

Note. It may be as well to verify independently the analytical conclusion immediately deducible from the preceding formulæ, viz. if λ, μ, ν, be given by the differential equations,

$$\kappa p = \frac{d\lambda}{dt} + \nu\frac{d\mu}{dt} - \mu\frac{d\nu}{dt},$$

$$\kappa q = -\nu\frac{d\lambda}{dt} + \frac{d\mu}{dt} + \lambda\frac{d\nu}{dt},$$

$$\kappa r = \mu\frac{d\lambda}{dt} - \lambda\frac{d\mu}{dt} + \frac{d\nu}{dt},$$

where $\kappa = 1 + \lambda^2 + \mu^2 + \nu^2$, and p, q, r, are any functions of t. Then if λ_0, μ_0, ν_0, be particular values of λ, μ, ν, and l, m, n, arbitrary constants, the general integrals are given by the system

$$P_0 = 1 - l\lambda_0 - m\mu_0 - n\nu_0,$$
$$P_0\lambda = l + \lambda_0 + m\nu_0 - n\mu_0,$$
$$P_0\mu = m + \mu_0 + n\lambda_0 - l\nu_0,$$
$$P_0\nu = n + \nu_0 + l\mu_0 - m\lambda_0.$$

Assuming these equations, we deduce the equivalent system,

$$(1+\lambda\lambda_0+\mu\mu_0+\nu\nu_0)\,l=\lambda-\lambda_0+\nu_0\mu-\nu\mu_0,$$

$$(1+\lambda\lambda_0+\mu\mu_0+\nu\nu_0)\,m=\mu-\mu_0+\lambda\nu-\lambda\nu_0,$$

$$(1+\lambda\lambda_0+\mu\mu_0+\nu\nu_0)\,n=\nu-\nu_0+\mu_0\lambda-\mu\lambda_0.$$

Differentiate the first of these and eliminate l, the result takes the form $0=$

$$\begin{aligned}&-(\mu_0^2+\nu_0^2)(\lambda'+\nu\mu'-\nu'\mu)-(\nu_0-\lambda_0\mu_0)(-\nu\lambda'+\mu'+\lambda\nu')+(\mu_0+\lambda_0\nu_0)(\mu\lambda'-\lambda\mu'+\nu')+\kappa_0\lambda'\\&+(\mu^2+\nu^2)(\lambda_0+\nu_0\mu_0-\nu_0'\mu_0)+(\nu-\lambda\mu)(-\nu_0\lambda_0+\mu_0'+\lambda\nu_0)-(\mu+\lambda\nu)(\mu_0\lambda-\lambda\mu_0'+\nu_0')-\kappa\lambda,\end{aligned}$$

where λ', &c. denote $\dfrac{d\lambda}{dt}$, &c. and $\kappa_0=1+\lambda_0^2+\mu_0^2+\nu_0^2$.

Reducing by the differential equations in λ, μ, ν; λ_0, μ_0, ν_0, this becomes

$$\kappa_0\{\lambda'+\tfrac{1}{2}p(\mu^2+\nu^2)+\tfrac{1}{2}q(\nu-\lambda\mu)-\tfrac{1}{2}r(\mu+\lambda\nu)\}$$

$$-\kappa\{\lambda_0+\tfrac{1}{2}p(\mu_0^2+\nu_0^2)+\tfrac{1}{2}q(\nu_0-\lambda_0\mu_0)-\tfrac{1}{2}r(\mu_0+\lambda_0\nu_0)\}=0;$$

or substituting for λ', λ_0, we have the identical equation

$$\tfrac{1}{2}p(\kappa_0\kappa-\kappa\kappa_0)=0,$$

and similarly may the remaining equations be verified.

7.

ON A CLASS OF DIFFERENTIAL EQUATIONS, AND ON THE LINES OF CURVATURE OF AN ELLIPSOID.

[*From the Cambridge Mathematical Journal*, vol. III. (1843), pp. 264–267.]

Consider the primitive equation

$$fx+gy+hz+\ldots=0, \tag{1}$$

between n variables x, y, z, the constants f, g, h being connected by the equation

$$H(f,g,h,\ldots)=0, \tag{2}$$

H denoting a homogeneous function. Suppose that f, g, h are determined by the conditions

$$\begin{gathered}fx_1+gy_1+hz_1+\ldots=0,\\ \vdots\\ fx_{n-1}+gy_{n-1}+hz_{n-1}+\ldots=0.\end{gathered} \tag{3}$$

Then writing

$$X=\begin{vmatrix}y, & z, & \ldots\\ y_1, & z_1, & \ldots\\ \vdots & & \\ y_{n-2}, & z_{n-2}, & \ldots\end{vmatrix}, \tag{4}$$

with analogous expressions for Y, Z, …; the equations (3) give f, g, h, … proportional to X, Y, Z, … or eliminating f, g, h, … by the equation (2),

$$H(X,Y,Z,\ldots)=0. \tag{5}$$

Conversely the equation (5), which contains, in appearance, $n(n-2)$ arbitrary constants, is equivalent to the system (1), (2). And if H be a rational integral function of the order r, the first side of the equation (5) is the product of r factors, each of them of the form given by the system (1), (2).

Now from the equation (1), we have the system

$$\begin{aligned} fx + gy + hz + \ldots &= 0, \\ fdx + gdy + hdz + \ldots &= 0, \\ &\vdots \\ fd^{n-2}x + gd^{n-2}y + hd^{n-2}z + \ldots &= 0, \end{aligned} \tag{6}$$

or writing

$$X = \begin{vmatrix} y, & z, & \ldots \\ dy, & dz, & \ldots \\ \vdots & & \\ d^{n-2}y, & d^{n-2}z, & \ldots \end{vmatrix}, \tag{7}$$

with analogous expressions for Y, Z, …; then from the equations (6), f, g, h are proportional to X, Y, Z, …: or, eliminating by (2),

$$H(X, Y, Z, \ldots) = 0. \tag{8}$$

Conversely the integral of the equation (8) of the order $(n-2)$ is given either by the system of equations (1), (2), in which it is evident that the number of arbitrary constants is reduced to $(n-2)$; or, by the equation (5), which contains in appearance $n(n-2)$ arbitrary constants, but which we have seen is equivalent in reality to the system (1), (2).

Thus, with three variables, the integral of

$$H(y\,dz - z\,dy,\ z\,dx - x\,dz,\ x\,dy - y\,dx) = 0 \tag{9}$$

may be expressed in the form

$$H(yz_1 - y_1z,\ zx_1 - z_1x,\ xy_1 - x_1y) = 0, \tag{10}$$

and also in the form

$$fx + gy + hz = 0, \tag{11}$$

where

$$H(f, g, h) = 0. \tag{12}$$

The above principles afford an elegant mode of integrating the differential equation for the lines of curvature of an ellipsoid. The equation in question is

$$(b^2 - c^2)\,x\,dy\,dz + (c^2 - a^2)\,y\,dz\,dx + (a^2 - b^2)\,z\,dx\,dy = 0, \tag{13}$$

where x, y, z are connected by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \tag{14}$$

writing

$$\frac{x^2}{a^2} = u, \qquad \frac{y^2}{b^2} = v, \qquad \frac{z^2}{c^2} = w, \tag{15}$$

these become

$$(b^2 - c^2)\,u\,dv\,dw + (c^2 - a^2)\,v\,dw\,du + (a^2 - b^2)\,w\,du\,dv = 0, \tag{16}$$

$$u + v + w = 1. \tag{17}$$

Multiplying by

$$-\{(v\,du - u\,dv)(w\,dv - v\,dw)(u\,dw - w\,du)\}^{-1},$$

the first of these becomes

$$\frac{-a^2\,du}{(v\,du - u\,dv)(u\,dw - w\,du)} + \frac{-b^2\,dv}{(w\,dv - v\,dw)(v\,du - u\,dv)} + \frac{-c^2\,dw}{(u\,dw - w\,du)(w\,dv - v\,dw)} = 0; \tag{18}$$

but writing (17) and its derived equations under the form

$$u + (v + w) = 1, \tag{19}$$

$$du + (dv + dw) = 0,$$

we deduce

$$-du(v + w) + u(dv + dw) = -du, \tag{20}$$

i.e.

$$-du = -(v\,du - u\,dv) + (u\,dw - w\,du), \tag{21}$$

and similarly

$$-dv = -(w\,dv - v\,dw) + (v\,du - u\,dv),$$

$$-dw = -(u\,dw - w\,du) + (w\,dv - v\,dw).$$

Substituting,

$$\frac{b^2 - c^2}{w\,dv - v\,dw} + \frac{c^2 - a^2}{u\,dw - w\,du} + \frac{a^2 - b^2}{v\,du - u\,dv} = 0; \tag{22}$$

the integral of which may be written in the form

$$\frac{b^2 - c^2}{wv_1 - vw_1} + \frac{c^2 - a^2}{uw_1 - wu_1} + \frac{a^2 - b^2}{vu_1 - uv_1} = 0, \tag{23}$$

where, on account of (17),

$$u_1 + v_1 + w_1 = 1; \tag{24}$$

and also in the form

$$fu + gv + hw = 0, \tag{25}$$

where f, g, h are connected by

$$\frac{f}{b^2 - c^2} + \frac{g}{c^2 - a^2} + \frac{h}{a^2 - b^2} = 0; \tag{26}$$

this last equation is satisfied identically by

$$f = \frac{B^2 - C^2}{b^2 - c^2}, \qquad g = \frac{C^2 - A^2}{c^2 - a^2}, \qquad h = \frac{A^2 - B^2}{a^2 - b^2}. \tag{27}$$

Restoring x, y, z, x_1, y_1, z_1 for u, v, w, u_1, v_1, w_1, the equations to a line of curvature passing through a given point x_1, y_1, z_1, on the ellipsoid, are the equation (14) and

$$\frac{(b^2 - c^2)}{a^2(y_1^2 z^2 - y^2 z_1^2)} + \frac{(c^2 - a^2)}{b^2(z_1^2 x^2 - z^2 x_1^2)} + \frac{(a^2 - b^2)}{c^2(x_1^2 y^2 - x^2 y_1^2)} = 0, \tag{28}$$

or again, under a known form, they are the equation (14) and

$$\frac{(b^2 - c^2)\,x^2}{B^2 - C^2} \cdot \frac{1}{a^2} + \frac{(c^2 - a^2)\,y^2}{C^2 - A^2} \cdot \frac{1}{b^2} + \frac{(a^2 - b^2)\,z^2}{A^2 - B^2} \cdot \frac{1}{c^2} = 0. \tag{29}$$

From the equations (14), (29) it is easy to prove the well-known form

$$\frac{x^2}{a^2 + \theta} + \frac{y^2}{b^2 + \theta} + \frac{z^2}{c^2 + \theta} = 1; \tag{30}$$

in fact, multiplying (29) by m, and adding to (14), we have the equation (30), if the equations

$$\begin{aligned} \frac{1}{a^2} + m\,\frac{1}{a^2}\,\frac{b^2 - c^2}{B^2 - C^2} &= \frac{1}{a^2 + \theta}, \\ \frac{1}{b^2} + m\,\frac{1}{b^2}\,\frac{c^2 - a^2}{C^2 - A^2} &= \frac{1}{b^2 + \theta},, \\ \frac{1}{c^2} + m\,\frac{1}{c^2}\,\frac{a^2 - b^2}{A^2 - B^2} &= \frac{1}{c^2 + \theta} \end{aligned} \tag{31}$$

are satisfied.

But on reduction, these take the form

$$\begin{aligned}(B^2-C^2)\,\theta+(b^2-c^2)\,m\,\theta+m\,a^2(b^2-c^2)&=0,\\(C^2-A^2)\,\theta+(c^2-a^2)\,m\,\theta+m\,b^2(c^2-a^2)&=0,\\(A^2-B^2)\,\theta+(a^2-b^2)\,m\,\theta+m\,c^2(a^2-b^2)&=0,\end{aligned}\tag{32}$$

and since, by adding, an identical equation is obtained, m and θ may be determined to satisfy these equations. The values of θ, m are

$$\theta=\frac{(a^2-b^2)(b^2-c^2)(c^2-a^2)}{a^2(B^2-C^2)+b^2(C^2-A^2)+c^2(A^2-B^2)},\tag{33}$$

$$m=\frac{b^2c^2(B^2-C^2)+c^2a^2(C^2-A^2)+a^2b^2(A^2-B^2)}{(a^2-b^2)(b^2-c^2)(c^2-a^2)}.\tag{34}$$

8.

ON LAGRANGE'S THEOREM.

[*From the Cambridge Mathematical Journal*, vol. III. (1843), pp. 283–286.]

THE value given by Lagrange's theorem for the expansion of any function of the quantity x, determined by the equation

$$x=u+hfx,\tag{1}$$

admits of being expressed in rather a remarkable symbolical form. The *à priori* deduction of this, independently of any expansion, presents some difficulties; I shall therefore content myself with showing that the form in question satisfies the equations

$$\frac{d}{du}\int F'xfx\,dx=\frac{d}{dh}\int F'x\,dx,\tag{2}$$

$$Fx=Fu \text{ for } h=0,\tag{3}$$

deduced from the equation (1), and which are sufficient to determine the expansion of Fx, considered as a function of u and h in powers of h.

Consider generally the symbolical expression

$$\phi\left(h\,\frac{d}{dh}\right)\Xi h,\tag{4}$$

$\phi\left(h\,\frac{d}{dh}\right)$ involving in general symbols of operation relative to any of the other variable entering into Ξh. Then, if Ξh be expansible in the form

$$\Xi h=\Sigma(Ah^m),\tag{5}$$

it is obvious that

$$\phi\left(h\,\frac{d}{dh}\right)\Xi h=\Sigma\{\phi(m)\,(Ah^m)\}=\Sigma\{(\phi(m)\,A)\,h^m\}.\tag{6}$$

For instance, m representing a variable contained in the function Ξh, and taking a particular form of $\phi\left(h\,\frac{d}{dh}\right)$,

$$\left(\frac{d}{du}\right)^k h^k\frac{d}{dh}\,\Xi h=\Sigma\left\{\left(\frac{d}{du}\right)^{km}Ah^m\right\}.\tag{7}$$

From this it is easy to demonstrate

$$\left(\frac{d}{du}\right)^k h^k\frac{d}{dh}\,\Xi h=\Sigma\left\{\left(\frac{d}{du}\right)^{km}A\,h^m\right\}.\tag{8}$$

where Πh denotes $\dfrac{d}{dh}\,\Xi h$, as usual. Hence also

$$\phi\Big(h\,\frac{d}{dh}\Big)\Xi h = \Sigma\bigg\{\Big(\frac{d}{du}\Big)^{km} A\,h^m\bigg\}. \tag{10}$$

of which a particular case is

$$\Big(\frac{d}{du}\Big)^{k} h^k\frac{d}{dh}\,(F'u\,e^{hfu}) = F'u\,(fu)^k\,e^{hfu}. \tag{11}$$

Also,

$$\Big(\frac{d}{du}\Big)^{0} h^0\frac{d}{dh}\,(F'u\,e^{hfu}) = Fu \text{ for } h = 0. \tag{12}$$

Hence the form in question for Fx is

$$Fx = \Big(\frac{d}{du}\Big)^{-1} h^{\frac{d}{dh}}\Big(F'u\,e^{hfu}\Big)\,; \tag{13}$$

from which, differentiating with respect to u, and writing F instead of F',

$$\frac{d}{du}Fx = \Big(\frac{d}{du}\Big)^{-1} h^{\frac{d}{dh}}\Big(Fu\,e^{hfu}\Big)\,, \tag{14}$$

a well-known form of Lagrange's theorem, almost equally important with the more usual one. It is easy to deduce (13) from (14). To do this, we have only to form the equation

$$F'x = \Big(\frac{d}{du}\Big)^{-1} h^{\frac{d}{dh}}\Big(F'uf'u\,e^{hfu}\Big)\,, \tag{15}$$

deduced from (14) by writing $Fxf'x$ for fx, and adding this to (14),

$$\begin{aligned} Fx &= \Big(\frac{d}{du}\Big)^{-1} h^{\frac{d}{dh}}\Big(Fu\,e^{hfu}\Big) - h\,\Big(\frac{d}{du}\Big)^{-1} h^{\frac{d}{dh}}\Big(Fuf'u\,e^{hfu}\Big) \\ &= \Big(\frac{d}{du}\Big)^{-1} h^{\frac{d}{dh}}\Big(F'u\,e^{hfu}\Big)\,. \end{aligned} \tag{16}$$

In the case of several variables, if

$$x = u + h\,f(x, x_1, \ldots), \qquad x_1 = u_1 + h_1 f_1(x, x_1, \ldots), \quad \text{\&c.}, \tag{17}$$

writing for shortness

$$F,\ f,\ f_1, \ldots \text{ for } F(u, u_1, \ldots),\ f(u, u_1, \ldots),\ f_1(u, u_1, \ldots),\ \ldots$$

then the formula is

$$\frac{F(x, x_1, \ldots)}{\{1 - hf'(x)\}\{1 - h_1 f_1'(x_1)\}\ldots} = \Big(\frac{d}{du}\Big)^{-1} h^{\frac{d}{dh}}\,\Big(\frac{d}{du_1}\Big)^{-1} h_1^{\frac{d}{dh_1}}\ldots(F\,e^{hf+h_1f_1+\cdots}), \tag{18}$$

(where $f'(x)$ is written to denote $\dfrac{d}{dx}f(x, x_1, \ldots)$, &c.)

Or the coefficient of $h^n h_1^{n_1}\ldots$ in the expansion of

$$\frac{F(x, x_1, \ldots)}{\{1 - hf'(x)\}\{1 - h_1 f_1'(x_1)\}\ldots} \tag{19}$$

is

$$\frac{1}{1\cdot 2\ldots n\cdot 1\cdot 2\ldots n_1\ldots}\,\Big(\frac{d}{du}\Big)^{n}\Big(\frac{d}{du_1}\Big)^{n_1}\ldots Ff^n f_1^{n_1}\ldots. \tag{20}$$

From the formula (18), a formula may be deduced for the expansion of $F(x, x_1, \ldots)$, in the same way as (13) was deduced from (14), but the result is not expressible in a simple form by this method. An apparently simple form has indeed been given for this expansion by Laplace, Mécanique Céleste, [Ed. 1, 1798] tom. I. p. 176; but the expression there given for the general term, requires first that certain differentiations should be performed, and then that certain changes should be made in the result, quantities x, x_1, ... are to be changed into x^n, $x_1^{n_1}$, ...; in other words, the general term is not really expressed by known symbols of operation only. The formula (18) is probably known, but I have not met with it anywhere.

9.

DEMONSTRATION OF PASCAL'S THEOREM.

[*From the Cambridge Mathematical Journal*, vol. IV. (1843), pp. 18–20.]

LEMMA 1. Let $U = Ax + By + Cz = 0$ be the equation to a plane passing through a given point taken for the origin, and consider the planes

$$U_1 = 0,\ U_2 = 0,\ U_3 = 0,\ U_4 = 0,\ U_5 = 0,\ U_6 = 0;$$

the condition which expresses that the intersections of the planes (1) and (2), (3) and (4), (5) and (6) lie in the same plane, may be written down under the form

$$\left|\begin{array}{c} A_1,\ \ A_2,\ \ A_3,\ \ A_4 \\ B_1,\ \ B_2,\ \ B_3,\ \ B_4 \\ C_1,\ \ C_2,\ \ C_3,\ \ C_4 \\ \\ A_3,\ A_4,\ A_5,\ A_6 \\ B_3,\ B_4,\ B_5,\ B_6 \\ C_3,\ C_4,\ C_5,\ C_6 \end{array}\right| = 0.$$

LEMMA 2. Representing the determinants

$$\left|\begin{array}{c} x_1, \\ y_1, \\ z_1 \end{array}\right|, \ \&\text{c}.$$

by the abbreviated notation $\overline{123}$, &c.; the following equation is identically true:

$$\overline{345}\cdot\overline{126} - \overline{346}\cdot\overline{125} + \overline{356}\cdot\overline{124} - \overline{456}\cdot\overline{123} = 0.$$

This is an immediate consequence of the equations

$$\left|\begin{array}{cccccc} x_1, & x_2, & x_3, & x_4, & x_5, & x_6 \\ y_1, & y_2, & y_3, & y_4, & y_5, & y_6 \\ z_1, & z_2, & z_3, & z_4, & z_5, & z_6 \\ x_1, & x_2, & x_3, & x_4, & x_5, & x_6 \\ y_1, & y_2, & y_3, & y_4, & y_5, & y_6 \\ z_1, & z_2, & z_3, & z_4, & z_5, & z_6 \end{array}\right| = 0.$$

Consider now the points 1, 2, 3, 4, 5, 6, the coordinates of these being respectively $x_1, y_1, z_1; \ldots x_6, y_6, z_6$. I represent, for shortness, the equation to the plane passing through the origin and the points 1, 2, which may be called the plane $\overline{12}$, in the form

$$x\,\overline{12}_x + y\,\overline{12}_y + z\,\overline{12}_z = 0;$$

consequently the symbols $\overline{12}_x$, $\overline{12}_y$, $\overline{12}_z$ denote respectively $y_1z_2 - y_2z_1$, $z_1x_2 - z_2x_1$, $x_1y_2 - x_2y_1$, and similarly for the planes $\overline{13}$, &c. If now the intersections of $\overline{12}$ and $\overline{45}$, $\overline{23}$ and $\overline{56}$, $\overline{34}$ and $\overline{61}$ lie in the same plane, we must have, by Lemma (1), the equation

$$\left|\begin{array}{cccc} \overline{12}_x, & \overline{45}_x, & \overline{23}_x, & \overline{56}_x \\ \overline{12}_y, & \overline{45}_y, & \overline{23}_y, & \overline{56}_y \\ \overline{12}_z, & \overline{45}_z, & \overline{23}_z, & \overline{56}_z \\ \overline{23}_x, & \overline{56}_x, & \overline{34}_x, & \overline{61}_x \\ \overline{23}_y, & \overline{56}_y, & \overline{34}_y, & \overline{61}_y \\ \overline{23}_z, & \overline{56}_z, & \overline{34}_z, & \overline{61}_z \end{array}\right| = 0.$$

Multiplying the two sides of this equation by the two sides respectively of the equation

$$\left|\begin{array}{ccc} x_5, & x_6, & x_1 \\ y_5, & y_6, & y_1 \\ z_5, & z_6, & z_1 \end{array}\right| = \overline{612}\cdot\overline{345},$$

and observing the equations

$$x_6\,\overline{12}_x + y_6\,\overline{12}_y + z_6\,\overline{12}_z = \overline{612}, \quad \overline{112} = 0, \ \&\text{c.}$$

this becomes

$$\overline{612}\left|\begin{array}{cccc} \overline{145}, & \overline{245}, & \dots & \dots \\ \overline{123}, & \dots & \dots & \overline{423} \\ \overline{156}, & \overline{256}, & \overline{356}, & \overline{456} \\ \dots & \overline{361}, & \dots & \overline{461} \\ \overline{523}, & \overline{534}, & \overline{561} & \end{array}\right| = 0;$$

reducible to

$$\overline{612}\cdot\overline{534}\left|\begin{array}{cccc} \overline{145}, & \overline{245}, & \dots & \\ \overline{123}, & \dots & \dots & \overline{423} \\ \overline{156}, & \overline{256}, & \overline{356}, & \overline{456} \\ \dots & \overline{361}, & \dots & \overline{461} \end{array}\right| = 0,$$

or, omitting the factor $\overline{612}\cdot\overline{534}$ and expanding,

$$\overline{145}\cdot\overline{256}\cdot\overline{423}\cdot\overline{361} + \overline{245}\cdot\overline{123}\cdot\overline{456}\cdot\overline{361} - \overline{245}\cdot\overline{123}\cdot\overline{356}\cdot\overline{461} - \overline{245}\cdot\overline{156}\cdot\overline{423}\cdot\overline{361} = 0.$$

Considering for instance x_6, y_6, z_6 as variable, this equation expresses evidently that the point 6 lies in a cone of the second order having the origin for its vertex, and the equation is evidently satisfied by writing x_6, y_6, $z_6 = x_1, y_1, z_1$, or x_3, y_3, z_3, or x_4, y_4, z_4, or x_5, y_5, z_5, and thus the cone passes through the points 1, 3, 4, 5. For x_6, y_6, $z_6 = x_2, y_2, z_2$, the equation becomes, reducing and dividing by $\overline{245}\cdot\overline{123}$,

$$\overline{452}\cdot\overline{321} - \overline{352}\cdot\overline{421} + \overline{152}\cdot\overline{423} = 0,$$

which is deducible from Lemma (2), by writing $x_6, y_6, z_6 = x_2, y_2, z_2$, and is therefore identically true. Hence the cone passes through the point (2), and therefore the points 1, 2, 3, 4, 5, 6 lie in the same cone of the second order, which is Pascal's Theorem. I have demonstrated it in the cone, for the sake of symmetry; but by writing throughout unity instead of z, the above applies directly to the case of the theorem in the plane.

The demonstration of Chasles' form of Pascal's Theorem (viz. that the anharmonic relation of the planes 61, 62, 63, 64 is the same with that of 51, 52, 53, 54), is very much simpler; but as it would require some preparatory information with reference to the analytical definition of the similarity of anharmonic relation, I must defer it to another opportunity.

10.

ON THE THEORY OF ALGEBRAIC CURVES.

[*From the Cambridge Mathematical Journal*, vol. IV. (1843), pp. 102–112.]

SUPPOSE a curve defined by the equation $U = 0$, U being a rational and integral function of the m^{th} order of the coordinates x, y. It may always be assumed, without loss of generality, that the terms involving x^m, y^m, both of them appear in U; and also that the coefficient of y^m is equal to unity: for in any particular curve where this was not the case, by transforming the axes, and dividing the new equation by the coefficient of y^m, the conditions in question would become satisfied. Let H_m denote the terms of U, which are of the order m, and let $y - \alpha x$, $y - \beta x$, $\ldots$ $y - \lambda x$ be the factors of H_m. If the quantities α, β, $\ldots$ λ are all of them different, the curve is said to have a number of asymptotic directions equal to the degree of its equation. Such curves only will be considered in the present paper, the consideration of the far more complicated theory of those curves, the number of whose asymptotic directions is less than the degree of their equation, being entirely rejected. Assuming, then, that the factors of H_m are all of them different, we may deduce from the equation $U = 0$, by known methods, the series

$$\begin{aligned} y &= \alpha x + \alpha' + \frac{\alpha''}{x} + \ldots, \\ y &= \beta x + \beta' + \frac{\beta''}{x} + \ldots, \\ &\vdots \\ y &= \lambda x + \lambda' + \frac{\lambda''}{x} + \ldots, \end{aligned} \tag{1}$$

and these being obtained, we have, identically,

$$U = (y - \alpha x - \alpha' - \ldots)(y - \beta x - \beta' - \ldots)\ldots(y - \lambda x - \lambda' - \ldots), \tag{2}$$

the negative powers of x on the second side, in point of fact, destroying each other.

Supposing in general that fx containing positive and negative powers of x, Efx denotes the function which is obtained by the rejection of the negative powers, we may write

$$U = E\left(y - \alpha x \ldots - \frac{\alpha^{m-1}}{x^{m-1}}\right)\left(y - \beta x \ldots - \frac{\beta^{m-1}}{x^{m-1}}\right)\ldots\left(y - \lambda x \ldots - \frac{\lambda^{m-1}}{x^{m-1}}\right), \tag{3}$$

the symbol E being necessary in the present case, because, when the series are continued only to the power x^{-m+1}, the negative powers no longer destroy each other.

We may henceforward consider U as originally given by the equation (3), the $m(m+1)$ quantities $\alpha, \alpha', \ldots \alpha^{m-1}, \beta, \beta', \ldots \beta^{m-1}, \ldots \lambda, \lambda', \ldots \lambda^{m-1}$ satisfying the equations obtained from the supposition that it is possible to determine the following terms $\alpha^{m+1}, \beta^{m+1}, \ldots \lambda^{m+1}, \ldots$ so that the terms containing negative powers of x, on the second side of equation (2), vanish. It is easily seen that $\alpha, \beta, \ldots \lambda, \alpha', \beta', \ldots \lambda'$ are entirely arbitrary, $\alpha'', \beta'', \ldots \lambda''$ satisfy a single equation involving only the preceding quantities, $\alpha''', \beta''', \ldots \lambda'''$ two equations involving the quantities which precede them, and so on, until $\alpha^{m-1}, \beta^{m-1}, \ldots \lambda^{m-1}$, which satisfy $(m-1)$ relations involving the preceding quantities. Thus the $m(m+1)$ quantities in question satisfy $\frac{1}{2}m(m-1)$ equations, or they may be considered as functions of $m(m+1) - \frac{1}{2}m(m-1) = \frac{1}{2}m(m+3)$ arbitrary constants. Hence the value of U, given by the equation (3), is the most general expression for a function of the m^{th} order. It is to be remarked also that the quantities $\alpha^{m+1}, \beta^{m+1}, \ldots \lambda^{m+1}$, are all of them completely determinable as functions of $\alpha, \beta, \ldots \lambda, \ldots \alpha^{m-1}, \beta^{m-1}, \ldots \lambda^{m-1}$.

The advantage of the above mode of expressing the function U, is the facility obtained by means of it for the elimination of the variable y from the equation $U = 0$, and any other analogous one $V = 0$. In fact, suppose V expressed in the same manner as U, or by the equation

$$V = E\left(y - Ax \ldots - \frac{A^{n-1}}{x^{n-1}}\right)\left(y - Bx \ldots - \frac{B^{n-1}}{x^{n-1}}\right)\ldots\left(y - Kx \ldots - \frac{K^{n-1}}{x^{n-1}}\right), \tag{4}$$

n being the degree of the function V. It is almost unnecessary to remark, that A, B, $\ldots$ K, $\ldots$ A^{n-1}, B^{n-1}, $\ldots$ K^{n-1} are to be considered as functions of $\frac{1}{2}n(n+3)$ arbitrary constants, and that the subsequent A^{n+1}, B^{n+1}, $\ldots$ K^{n+1}, $\ldots$ can be completely determined as functions of these. Determining the values of

y from the equation (3), viz. the values given by the equations (2); substituting these successively in the equation

$$V = (y - Ax - \ldots)(y - Bx - \ldots)\ldots(y - Kx - \ldots) = 0, \tag{5}$$

analogous to (2), and taking the product of the quantities so obtained, also observing that this product must be independent of negative powers of x, the result of the elimination may be written down under the form

$$E\big[\{(\alpha - A)x\ldots + \tfrac{1}{x^{mn-1}}\}\ldots\{(\alpha - K)x\ldots + \tfrac{1}{x^{mn-1}}\}\ldots \times \{(\lambda - A)x\ldots + \tfrac{1}{x^{mn-1}}\}\ldots\{(\lambda - K)x\ldots + \tfrac{1}{x^{mn-1}}\}\big], \tag{6}$$

the series in { } being continued only to x^{-mn+1}, because the terms after this point produce in the whole product only terms involving negative powers of x. It is for the same reason that the series in ( ), in the equations (3) and (4), are only continued to the terms involving x^{-m+1}, x^{-n+1} respectively.

The first side of the equation (6) is of the order mn, in x, as it ought to be. But it is easy to see, from the form of the expression, in what case the order of the first side reduces itself to a number less than mn. Thus, if n be not greater than m, and the following equations be satisfied,

$$\begin{gathered} A = \alpha, \quad A^{(1)} = \alpha^{(1)}, \quad \ldots \quad A^{r-1} = \alpha^{r-1}, \quad r \not> n, \\ \ldots \quad B = \beta, \quad B^{(1)} = \beta^{(1)}, \quad \ldots \quad B^{s-1} = \beta^{s-1}, \quad s \not> n, \\ \vdots \end{gathered} \tag{7}$$

the degree of the equation (6) is evidently $mn - r - s \ldots - v$, or the curves $U = 0$, $V = 0$ intersect in this number only of points. If $mn - r - s \ldots - v = 0$, the curves $U = 0$ and $V = 0$ do not intersect at all, and if $mn - r - s \ldots - v$ be negative, $= -\omega$ suppose, the equation (6) is satisfied identically; or the functions U, V have a common factor, the number ω expressing the degree of this factor in x, y.

Supposing the function V given arbitrarily, it may be required to determine U, so that the curves $U = 0$, $V = 0$ intersect in a number $mn - k$ points. This may in general be done, and done in a variety of ways, for any value of k from unity to $\frac{1}{2}m(m + 3)$. I shall not discuss the question generally at present, nor examine into the meaning of the quantity $mn - \frac{1}{2}m(m + 3)$ $\{= \frac{1}{2}m(2n - m - 3)\}$ becoming negative, but confine myself to the simple case of U and V, both of them functions of the second order. It is required, then, to find the equations of all those curves of the second order which intersect a given curve of the second order in a number of points less than four.

Assume in general

$$V = E(y - Ax - A')\left(y - Bx - B' - \frac{B''}{x}\right),$$

then A'', B'' satisfy $A'' + B'' = 0$, and putting $B'' = \dfrac{K}{A - B}$, and therefore $A'' = -\dfrac{K}{A - B}$, and reducing, we obtain

$$V = (y - Ax - A')(y - Bx - B') + K.$$

Similarly assume $U = E(y - \alpha x - \alpha')\left(y - \beta x - \beta' - \dfrac{\beta''}{x}\right)$, then α'', β'' satisfy $\alpha'' + \beta'' = 0$, and putting $\beta'' = \dfrac{k}{\alpha - \beta}$, and therefore $\alpha'' = -\dfrac{k}{\alpha - \beta}$, and reducing, we obtain

$$U = (y - \alpha x - \alpha')(y - \beta x - \beta') + k.$$

Suppose

(1) $U = 0$, $V = 0$ intersect in three points, we must have $\alpha = A$, or the curve $U = 0$ must have one of its asymptotes parallel to one of the asymptotes of $V = 0$.

(2) The curves intersect in two points. We must have $\alpha = A$, $\alpha' = A'$, or else $\alpha = A$, $\beta = B$; i.e. $U = 0$ must have one of its asymptotes coincident with one of the asymptotes of the curve $V = 0$, or else it must have its two asymptotes parallel to those of $V = 0$. The latter case is that of similar and similarly situated curves.

(3) Suppose the curves intersect in a single point only. Then either $\alpha = A$, $\alpha' = A'$, $\alpha'' = A''$, which it is easy to see gives

$$U = (y - Ax - A')(y - \beta x - \beta') + K\,\frac{\alpha - \beta}{A - B},$$

or else $\alpha = A$, $\alpha' = A'$, $\beta = B$, which is the case of one of the asymptotes of the curve $U = 0$, coinciding with one of those of the curve $V = 0$, and the remaining asymptotes parallel. As for the first case, if Ω, Ω_1 are the transverse axes, θ, θ_1 the inclinations of the two asymptotes to each other, these four quantities are connected by the equation

$$\frac{\Omega^2}{\Omega_1^2} = \frac{\tan\frac{1}{2}\theta\,\cos^2\frac{1}{2}\theta}{\tan\frac{1}{2}\theta_1\,\cos^2\frac{1}{2}\theta_1},$$

and besides, one of the asymptotes of the first curve is coincident with one of the asymptotes of the second. This is a remarkable case; it may be as well to verify that $U = 0$, $V = 0$ do intersect in a single point only. Multiplying the first equation by $y - Bx - B'$, the second by $y - \beta x - \beta'$, and subtracting, the result is

$$(A - \beta)(y - Bx - B') - (A - B)(y - \beta x - \beta') = 0,$$

reducible to

$$y - Ax = \frac{A(B' - \beta') + B\beta' - B'\beta}{A - B}\,\frac{1}{A - \beta}, \quad \text{i.e. } y - Ax - C = 0.$$

Combining this with $V = 0$, we have an equation of the form $y - Bx - D = 0$. And from this and $y - Ax - C = 0$, x, y are determined by means of a simple equation.

(4) Lastly, when the curves do not intersect at all. Here $\alpha = A$, $\alpha' = A'$, $\beta = B$, $\beta' = B'$, or the asymptotes of $U = 0$ coincide with those of $V = 0$; i.e. the curves are similar, similarly situated, and concentric: or else $\alpha = A$, $\alpha' = A'$, $\alpha'' = A''$, $\beta = B$; here

$$U = (y - Ax - A')(y - Bx - B') + K,$$

or the required curve has one of its asymptotes coincident with one of those of the proposed curve; the remaining two asymptotes are parallel, and the magnitudes of the curves are equal.

In general, if two curves of the orders m and n, respectively, are such that r asymptotes of the first are parallel to as many of the second, s out of these asymptotes being coincident in the two curves, the number of points of intersection is $mn - r - s$; but the converse of this theorem is not true.

In a former paper, *On the Intersection of Curves*, [5], I investigated the number of arbitrary constants in the equation of a curve of a given order p subjected to pass through the mn points of intersection of two curves of the orders m and n respectively. The reasoning there employed is not applicable to the case where the two curves intersect in a number of points less than mn. In fact, it was assumed that, $W = 0$ being the equation of the required curve, W was of the form $uU + vV$; u, v being polynomials of the degrees $p - m$, $p - n$ respectively. This is, in point of fact, true in the case there considered, viz. that in which the two curves intersect in mn points; but where the number of points of intersection is less than this, u, v may be assumed polynomials of an order higher than $p - m$, $p - n$, and yet $uU + vV$ reduce itself to the order p. The preceding investigations enable us to resolve the question for every possible case.

Considering then the functions U, V determined as before by the equations (3), (4), suppose, in the first place, we have a system of equations

$$\alpha = A,\ \beta = B,\ \ldots\ \theta = H \quad (t \text{ equations}). \tag{8}$$

Assume

$$\begin{aligned} P &= (y - \alpha x - \ldots)(y - \beta x - \ldots)\ldots(y - \theta x - \ldots), \\ Q &= (y - Ax - \ldots)(y - Bx - \ldots)\ldots(y - Hx - \ldots), \\ T &= (y - \iota x - \ldots)\ldots(y - \lambda x - \ldots), \\ \Psi &= (y - Ix - \ldots)\ldots(y - Kx - \ldots); \end{aligned}$$

whence $U = PT$, $V = Q\Psi$.

Suppose $T = ET + \Delta T$, $\Psi = E\Psi + \Delta\Psi$,

$$\begin{aligned} E\Psi \cdot U - ET \cdot V &= E\Psi \cdot PT - ET \cdot Q\Psi, \\ &= E\Psi \cdot P(ET + \Delta T) - ET \cdot Q(E\Psi + \Delta\Psi), \\ &= ET \cdot E\Psi \cdot (P - Q) + E\Psi \cdot P \cdot \Delta T - ET \cdot Q \cdot \Delta\Psi, \\ &= E[ET \cdot E\Psi \cdot (P - Q) + E\Psi \cdot P\Delta T - ET \cdot Q \cdot \Delta\Psi], \\ &= \Pi \text{ suppose.} \end{aligned}$$

In this expression ET, $E\Psi$ are of the degrees $m-t$, $n-t$, ΔT, $\Delta\Psi$ of the degree -1, and P, Q, $P-Q$ of the degrees t, t, $t-1$ respectively. The terms of Π are therefore of the degrees $m+n-t-1$, $m-1$, $n-1$ respectively, and the largest of these is in general $m+n-t-1$. Suppose, however, that $m+n-t-1$ is equal to $m-1$ (it cannot be inferior to it), then $t = n$; Ψ becomes equal to unity, or $\Delta\Psi$ vanishes. The remaining two terms of Π are $ET(P-Q)$, $P\Delta T$, which are of the degrees $m-1$, $n-1$ respectively. Π is still of the degree $m-1$, supposing $m > n$. If $m = n$, the term $P\Delta T$ vanishes. Π is still of the degree $m-1$. Hence in every case the degree of Π is $m+n-t-1$: assuming always that $P-Q$ does not reduce itself to a degree lower than $t-1$, (which is always the case as long as the equations $\alpha' = A'$, $\beta' = B'$, $\ldots$ $\theta' = H'$ are not all of them satisfied simultaneously). It will be seen presently that we shall gain in symmetry by wording the theorem thus: the degree of Π is equal to the greatest of the two quantities $m+n-t-1$, $m-1$.

Suppose next, in addition to the equations (8), we have

$$\alpha' = A', \ \beta' = B', \ \ldots \ \theta' = F', \quad t' \text{ equations}, \ t' \not> t. \tag{8'}$$

Then, taking T', Ψ', P', Q' the analogous quantities to T, Ψ, P, Q, and putting

$$E\Psi' \cdot U - ET' \cdot V = \Pi',$$

we have, as before,

$$\Pi' = E\{ET \cdot E\Psi' \cdot (P' - Q') + E\Psi' \cdot P'\Delta T' - ET \cdot Q'\Delta\Psi'\}.$$

The degree of $P' - Q'$ is $t' - 2$ (unless simultaneously $\alpha'' = A''$, $\beta'' = B''$, $\ldots$ $\theta'' = F''$, in which case the degree may be lower). The degrees, therefore, of the terms of Π' are $m+n-t'-2$, $n-1$, $m-1$. Or we may say that the degree of Π' is equal to the greatest of the quantities $m+n-t'-2$, $m-1$; though to establish this proposition in the case where $t' = n-1$ would require some additional considerations.

Continuing in this manner until we come to the quantity $\Pi^{(k-1)}$, the degree of this quantity is the greatest of the two numbers $m+n-t^{(k-1)}-k$, $m-1$. And we may suppose that none of the equations $\alpha^k = A^{(k)}$, $\ldots$ are satisfied, so that the series $\Pi, \Pi', \ldots \Pi^{(k-1)}$ is not to be continued beyond this point.

Considering now the equation of the curve passing through the $mn-t-t' \ldots -t^{(k-1)}$ points of intersection of $U = 0$, $V = 0$, we may write

$$W = uU + vV + p\Pi + p'\Pi' + \ldots + p^{(k-1)}\Pi^{(k-1)} = 0, \tag{9}$$

for the required equation; the dimensions of u, v, p, p', $\ldots p^{(k-1)}$ being respectively

$$p - m, \quad p - n; \qquad p - m - n + t + 1 \text{ or } p - m + 1;$$

$$p - m - n + t' + 2 \text{ or } p - m + 1; \quad \ldots \quad p - m - n + t^{(k-1)} + k \text{ or } p - m + 1,$$

the lowest of the two numbers being taken for the dimensions of p, p', $\ldots p^{(k-1)}$. Also, if any of these numbers become negative, the corresponding term is to be rejected. In saying that the degrees of p, p', $\ldots p^{(k-1)}$ have these actual values, it is supposed that the degrees of Π, Π', $\ldots \Pi^{(k-1)}$ actually ascend to the greatest of the values

$$p - m - n + t + 1, \text{ or } m - 1; \quad m + n - t' - 2, \text{ or } m - 1; \quad \ldots \quad m + n - t^{(k-1)} + k, \text{ or } m - 1.$$

The cases of exception to this are when several of the consecutive numbers $t, t', \ldots t^{(k-1)}$ are equal. In this case the corresponding terms of the series $\Pi, \Pi', \ldots \Pi^{(k-1)}$, are also equal. Suppose for instance t, t'

were equal, Π, Π' would also be equal. A term of p of an order higher by unity than $p-m-n+t+1$, or $p-m+1$, which is the highest term admissible, produces in $p\Pi$ a term identical with one of the terms of $p'\Pi$; so that nothing is gained in generality by admitting such terms into p. The equation (9), with the preceding values for the dimensions of p, p', $\dots p^{(k-1)}$, may be employed, therefore, even when several consecutive terms of the series $t, t', \dots t^{(k-1)}$ are equal. It will be convenient also to assume that $p-m$ is not negative, or at least for greater simplicity to examine this case in the first place.

uU, and vV, contain terms of the form $x^\alpha y^\beta U$, $x^\gamma y^\delta V$, $\alpha+\beta \le p-m$, $\gamma+\delta \le p-n$; $p\Pi$ contains terms of this form, and in addition terms for which $\alpha+\beta = p+1-m$, $\gamma+\delta = p+1-n$. It is useless to repeat the former terms, so that we may assume for p, a homogeneous function of the order $p-m-n+t+1$, or $p-m+1$; in which case $p\Pi$ consists only of terms for which $\alpha+\beta = p+1-m$, $\gamma+\delta = p+1-n$. And the general expression of p contains $p-m-n+t+2$, or $p-m+2$, arbitrary constants. Similarly $p'\Pi'$ contains terms of the form of those in uU, vV, $p\Pi$, and also terms for which $\alpha+\beta = p+2-m$, $\gamma+\delta = p+2-n$; the latter terms only need be considered, or p' may be assumed to be a homogeneous function of the order $p-m-n+t'+2$, or $p-m+1$, containing therefore $p-m-n+t'+3$, or $p-m+2$ arbitrary constants. Similarly $p^{(k-1)}$ contains $p-m-n+t^{(k-1)}+k+1$ or $p-m+2$ arbitrary constants. Assume

$$\nu = \binom{p-m-n+t+2}{p-m+2} + \binom{p-m-n+t'+3}{p-m+2} \dots + \binom{p-m-n+t^{(k-1)}+k+1}{p-m+2} \dots, \qquad (10)$$

where, in forming the value of ν the least of the two quantities in ( ) is to be taken; this value also, if negative, being replaced by zero. The number of arbitrary constants in $p, p', \dots p^{(k-1)}$ is consequently equal to ν.

The numbers of arbitrary constants in u, v, are respectively

$$\{1+2+\dots+(p-m+1)\} \quad \text{and} \quad \{1+2+\dots+(p-n+1)\}$$

i.e. $\frac{1}{2}(p-m+1)(p-m+2)$, and $\frac{1}{2}(p-n+1)(p-n+2)$; thus the whole number of arbitrary constants in W, diminished by unity (since nothing is gained in generality, by leaving the coefficient (for instance of y) indeterminate, instead of supposing it equal to unity) becomes

$$\tfrac{1}{2}(p-m+1)(p-m+2) + \tfrac{1}{2}(p-n+1)(p-n+2) + \nu - 1,$$

reducible to

$$\tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+1)(p-m-n+2) - mn + \nu.$$

By the reasonings contained in the paper already referred to, if $p+k-m-n+1$ be positive, to find the number of really disposable constants in W, we must subtract from this number a number $\frac{1}{2}(p+k-m-n+1)(p+k-m-n+2)$. Hence, calling ϕ the number of disposable constants in W, we have

$$\phi = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+1)(p-m-n+2) - mn + \nu - \Lambda, \qquad (11)$$

where

$$\Lambda = 0, \text{ if } p+k-m-n+1 \text{ be negative or zero}, \qquad (12)$$

$$\Lambda = \tfrac{1}{2}(p+k-m-n+1)(p+k-m-n+2),$$

if $p+k-m-n+1$ be positive; and ν is given by the equation (9).

Also, if θ be the number of points through which the curve $W=0$ can be drawn, including the points of intersection of the curves $U=0$, $V=0$, then

$$\theta = \phi + (mn - t - t' \dots - t^{(k-1)}) \quad \text{or}$$

$$\theta = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+1)(p-m-n+2) + \nu - \Lambda - t - t' \dots - t^{(k-1)}. \qquad (13)$$

Any particular cases may be deduced with the greatest facility from these general formulæ. Thus, supposing the curves to intersect in the complete number of points mn, we have

$$\phi = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(1-\epsilon)(p-m-n+1)(p-m-n+2) - mn,$$

ϵ being zero or unity according as $p < m+n-1$ or $p > m+n-1$. Reducing, we have, for $p \not> m+n-3$,

$$\phi = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+1)(p-m-n+2) - mn,$$

$$\theta = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+1)(p-m-n+2);$$

and for $p > m+n-3$,

$$\phi = \tfrac{1}{2}p(p+3) - mn,$$

$$\theta = \tfrac{1}{2}p(p+3).$$

Suppose, in the next place, the curves have t parallel pairs of asymptotes, none of these pairs being coincident. Then

$$p \not> m+n-t-2,$$

$$\phi = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+1)(p-m-n+2) - mn,$$

$$\theta = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+1)(p-m-n+2) - t;$$

$$p > m+n-t-2, \quad p \not> m+n-2,$$

$$\phi = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+2)(p-m-n+3) - mn + t,$$

$$\theta = \tfrac{1}{2}p(p+3) + \tfrac{1}{2}(p-m-n+2)(p-m-n+3);$$

$$p > m+n-2,$$

$$\phi = \tfrac{1}{2}p(p+3) - mn + t,$$

$$\theta = \tfrac{1}{2}p(p+3).$$

In these formulæ, if t be equal to 2 or greater than 2, the limiting conditions are more conveniently written

$$p \not> m+n-t-2; \quad p \not> m+n-t-2 > m+n-4; \quad p > m+n-4.$$

Similarly may the solution of the question be explicitly obtained when the curves have t asymptotes parallel, and θ out of these coincident, but the number of separate formulæ will be greater.

In conclusion, I may add the following references to two memoirs on the present subject: the conclusions in one point of view are considerably less general even than those of my former paper, though much more so in another. Jacobi, *Theoremata de punctis intersectionis duarum curvarum algebraicarum*; Crelle's Journal, vol. XV. [1836, pp. 285–308]; Plücker, *Théorèmes généraux concernant les équations à plusieurs variables, d'un degré quelconque entre un nombre quelconque d'inconnues*, *Do* vol. XVI. [1837, pp. 47–57].

[Paper 10 continues on p. 76.]